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STATISTICAL ANALYSIS PLAN

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LEVOECMO

LEVOSIMENDAN to facilitate weaning from ECMO in severe cardiogenic shock patients

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Sommaire

1 Summary of the study

LEVOECMO, multicenter, randomized, double-blind, phase III clinical trial investigates whether the early administration of levosimendan can reduce the duration of ECMO (Extracorporeal Membrane Oxygenation) support in patients with severe cardiogenic shock. Reducing ECMO duration may decrease ECMO-related complications such as hemorrhage, stroke, and infections, as well as shorten ICU and hospital stays, ultimately lowering healthcare costs for these critically ill patients. Additionally, earlier ECMO weaning may lead to improved long-term outcomes. This trial aims to demonstrate the superiority of levosimendan. The enrolment period will span 24 months, with each subject participating for a total of 60 days, including a 24-hour treatment period and 60-day follow-up.

2 Objectives and endpoints

2.1 Main objective and primary endpoint

The primary objective is to assess the efficacy of early administration of levosimendan on the time to successful ECMO weaning in patients receiving VA-ECMO support for acute cardiogenic shock refractory to conventional therapy.

The primary endpoint will be time to successful ECMO weaning within the 30 days following randomization.

- The successful weaning of ECMO (within 30 days after randomization) is defined as the patient surviving without ECMO, other mechanical circulatory support device or heart transplantation 30 days after ECMO removal. Thus, all ECMO weaning from randomization to 30 days after randomization will be considered, and the qualification for successful ECMO weaning will need 30 days of follow-up after ECMO removal (thus until day 60 after randomization for an ECMO weaning performed on day 30 after randomization).
- ECMO weaning failure (i.e., unsuccessful weaning), within 30 days after randomization, is a competing event.
- Death within 30 days after randomization, without an attempt to wean, is a competing event.
- No event after 30 days after randomization will be considered and all patients without any of the three events defined above are censored at 30 days post randomization.

2.2 Secondary endpoints

The secondary endpoints of the trial include:

- **Mortality**: Mortality rates at day 30 (D30) and day 60 (D60).
- **ECMO Support Duration**: Total duration of ECMO support between inclusion and D30, and between inclusion and D60; and the number of ECMO-free days from inclusion to D30/D60.
- **Duration of ICU and Hospital Stay**: Length of stay in the ICU and total hospitalization duration.
- **Major Adverse Cardiovascular Events (MACE)**: Defined as the occurrence, on days 30 and 60, of any of the following: death, cardiac transplant, escalation to a permanent left ventricular assist device, stroke, or re-hospitalization for heart failure.
- **Hemodynamic Improvement**: Time to improvement in hemodynamic parameters defined as achieving a mean blood pressure > 60 mmHg.
- **Organ Failure**: Number of days with organ failure, defined as a Sequential Organ Failure Assessment (SOFA) score > 2 in at least one of the six organ systems, and the number of organ failure-free days between inclusion and D30.
- **Hemodynamic Support with Catecholamines**: Duration of catecholamine support and the number of catecholamine-free days between inclusion and D30.
- **Mechanical Ventilation**: Duration of mechanical ventilation and the number of ventilatory-free days between inclusion and D30, as well as between inclusion and D60.
- **Renal replacement therapy (RRT)**: Duration of RRT and the number of RRT-free days between inclusion and D30.
- **Left Ventricular Function**: Left ventricular function assessed via echocardiography on day 30 post-randomization by Fraction Ejection of Left Ventricular (FELV) in %.
- **Safety:**
    1. Drug Adverse Effects: Incidence of drug-related adverse effects, including atrial fibrillation, ventricular arrhythmias, ventricular fibrillation, torsades de pointes, and hypokalemia.
    2. Any adverse events observed during the follow-up period

3 Methods

3.1 Flow chart

A flow chart will be included to illustrate the flow of participants throughout the study. This chart will display: the number of eligible patients screened for inclusion, the total number of patients effectively included and randomized across both treatment arms, the distribution of participants in each arm, patient follow-up details, including the number of patients lost to follow-up in each arm and the number of patients analyzed in each group.

3.2 Assumptions for calculating the number of subjects required

The cumulative incidence of successful ECMO weaning in the control arm was estimated at 50%, based on a 30-day cohort study of similar patients, considering competing risks (death or weaning failure). According to Latouche et al [1], with a subdistribution hazard ratio (HR) of 1.75, a two-sided alpha risk of 5%, and a power of 80%, a total of 101 events (successful ECMO weaning) are required. This corresponds to a total sample size of 206 subjects, with 103 patients per group.

3.3 Analysis population

The primary analysis will be conducted on an intention-to-treat (ITT). All randomized patients will be included in the analysis in their originally assigned groups. All patient will be analyzed, regardless of any treatment changes that occur over the course of the study.

3.4 Descriptive analysis

Qualitative variables will be reported by frequencies and percentages. Quantitative variables will be summarized based on the distribution of the data: for normally distributed variables, results will be expressed as mean and standard deviation (SD); for non-normally distributed variables, results will be expressed as median and interquartile range (IQR), along with range values. For censored data, descriptive statistics will include Kaplan-Meier estimates or cumulative incidence functions, providing rates at 15, 30, and 60 days with corresponding 95% confidence intervals where appropriate.

3.5 Primary endpoint analysis

The primary outcome is the time to successful ECMO weaning within the 30 days following randomization, accounting for competing risks.

Definition of Successful ECMO Weaning:

- Successful ECMO weaning (within 30 days post-randomization) is defined as the patient surviving without ECMO, additional mechanical circulatory support, or heart transplantation for 30 days following ECMO removal.

Types of Follow-Up Termination within 30 Days:

- **Successful ECMO weaning**: Event of interest
- **Not weaned from ECMO at 30 days**: Censored
- **Death before day 30 without ECMO weaning**: Competing event
- **Unsuccessful ECMO weaning** (i.e., death, need for repeat ECMO, additional mechanical circulatory support, or heart transplantation within 30 days post-ECMO removal): Competing event
- **Lost to follow-up within 30 days**: Censored

Definitions of Follow-Up Times:

- **Time to Successful ECMO Weaning**: Interval between the date of randomization and the date of successful ECMO weaning.
- **Time to Unsuccessful ECMO Weaning**: Interval between the date of randomization and the ECMO removal date, for patients who experience death, repeat ECMO, need for alternative mechanical support, or heart transplantation within 30 days after ECMO removal.
- **Time to Death without ECMO Weaning**: Interval between the date of randomization and the date of death.
- **Time to Lost to Follow-Up**: Interval between the date of randomization and the date of last known new.

Subjects without any of the above events by 30 days will be censored, with time recorded as 30 days.

Analysis:

- This outcome will be summarized using the **Cumulative Incidence Function (CIF)**, providing rates at 15 and 30 days with 95% confidence intervals. Additionally, cumulative incidence functions will be plotted for each treatment group. The two CIF for successful ECMO Weaning (event of interest) in the two treatment groups will be compared using a Gray's test.
- A Fine and Gray model will be performed to account for competing risks, **adjusted on the primary etiology of cardiogenic shock** (acute myocardial infarction, myocarditis, post-heart surgery and other causes). It will estimate **subdistribution Hazard Ratios (sHRs)** with 95% confidence intervals for the treatment group and for each event defined above.
- The proportional hazards assumption will be tested using Schoenfeld residuals.
- A sensitivity analysis will be performed using a Cause-specific Cox model. **Cause-specific Hazard Ratios (cHRs)** with 95% confidence intervals for the treatment group will be estimated.

3.6 Secondary endpoints analyses

- **30-Day and 60-Day Mortality**: It will be described by survival analysis with Kaplan-Meier estimation and 95% confidence interval. Only survival analysis within 60 days will be performed. In this analysis, time to death will be defined as the difference between the date of randomization and the date of death (whatever ECMO weaning had occurred before or not) or the date of the last news or the end of the follow-up. Difference between groups will be tested with the logrank test. To adjust for the primary etiology of cardiogenic shock, a cox model will be used.
- **ECMO Support Duration**: Total duration of ECMO support between inclusion and D30, and between inclusion and D60; and the number of ECMO-free days from inclusion to D30/D60 will be compared using the Wilcoxon rank sum test, with deaths imputed as 0 ECMO-free days. To adjust for the primary etiology of cardiogenic shock, a general linear model will be used.
- **Duration of ICU and Hospital Stay**: Duration of ICU stay and total hospital stay will be described using restricted mean survival times (RMST) up to day 60, with corresponding 95% confidence intervals. An analysis of time-to-event data is required because some patients may still be hospitalized (in ICU or in the hospital) at the end of the 60-day follow-up period, leading to right-censoring. Group comparisons will be performed using the log-rank test.. To adjust for the primary etiology of cardiogenic shock, a Cox proportional hazards model will be used.
- **Major Adverse Cardiovascular Events (MACE)**: MACE is defined as the occurrence of any of the following events on days 30 and 60: death, cardiac transplant, escalation to a permanent left ventricular assist device, stroke, or re-hospitalization for heart failure. The incidence of MACE will be described using survival analysis with Kaplan-Meier estimation and its 95% confidence interval. Only survival analysis up to 60 days will be performed. In this analysis, time will be defined as the time from the date of randomization and the date of MACE occurrence, the date of the last follow-up update, or the end of follow-up, whichever comes first. Differences between groups will be assessed using the log-rank test. To adjust for the primary etiology of cardiogenic shock, a Cox proportional hazards model will be employed.
- **Hemodynamic Improvement (revised)**: Time to improvement in hemodynamic parameters, initially defined as achieving a mean blood pressure (MBP) > 60 mmHg, was pre-specified for analysis using a competing risks framework. However, after inspection of the descriptive data and prior to conducting any inferential analysis, it was observed that most patients maintained a stable hemodynamic profile (MBP > 60 mmHg) throughout follow-up. As a result, this endpoint was deemed non-informative and removed from inferential analyses. A descriptive graphical representation of MBP evolution up to Day 30 was provided instead.
- **Organ Failure**: Number of days with organ failure, as defined by the SOFA (Sequential Organ Failure Assessment) score > 2 in at least one of the six organ systems, and the number of organ failure-free days between inclusion and D30, will be compared using the Wilcoxon rank sum test, with deaths imputed as 0 failure-free days. To adjust for the primary etiology of cardiogenic shock, a general linear model will be used.

- **Hemodynamic Support with Catecholamines**: Duration of catecholamine support and the number of catecholamine-free days between inclusion and D30, will be compared using the Wilcoxon rank sum test, with deaths imputed as 0 catecholamines-free days. To adjust for the primary etiology of cardiogenic shock, a general linear model will be used.
- **Mechanical Ventilation**: Duration of mechanical ventilation and the number of ventilatory-free days between inclusion and D30, as well as between inclusion and D60, will be compared using the Wilcoxon rank sum test, with deaths imputed as 0 ventilatory-free days. To adjust for the primary etiology of cardiogenic shock, a general linear model will be used.
- **Renal replacement therapy (RRT)**: Duration of RRT and the number of RRT-free days between inclusion and D30 will be compared using the Wilcoxon rank sum test, with deaths imputed as 0 ventilatory-free days. To adjust for the primary etiology of cardiogenic shock, a general linear model will be used.
- **Left Ventricular Function**: To compare the two treatment groups, we will use the Global Net Benefit method at Day 30, base on a hierarchical composite event:
    1. Death
    2. Heart transplantation
    3. Left ventricular function improvement defined as achieving a left ventricular ejection fraction (LVEF) > 30%

  Each patient in the experimental group will be pairwise compared to each patient in the control group:

    - A patient from the experimental group is considered a "win" if he survives while the matched control patient dies.
    - If both patients survive, a "win" occurs if the experimental group patient does not require heart transplantation while the control group patient does.
    - If neither patient undergoes transplantation, the patient with greater left ventricular function improvement at Day 30 is considered the winner.
    - If both patients have the same outcome at each hierarchical level, the comparison is classified as a "tie".

  The Global Net Benefit will be: Number of Wins−Number of Losses/Total number of comparisons.

  A positive Net Benefit value indicates a favorable effect of the experimental treatment compared to the control and 95% Confidence interval of the Net Benefit will be estimated using bootstrap.

- **Safety**:
    1. Adverse drug reactions: The incidence of drug-related adverse effects (e.g., atrial fibrillation, ventricular arrhythmias, ventricular fibrillation, torsades de pointes, hypokalemia) will be compared between the two treatment groups using Poisson regression model considering that a patient may experiment more than one event. The model will be adjusted for prognostic factors.

2. Any adverse events observed during the follow-up period will be described overall and according to their nature by treatment group, in terms of number, and percentage of patients affected. Serious adverse events will be described separately. The number of adverse events and the number of serious adverse events will be compared between the two groups using Poisson regression model adjusted for the primary etiology of cardiogenic shock.

3.7 Statistical significance level planned

All statistical analyses will be conducted using a two-sided alpha level of 5%.

3.8 Missing data management

For the primary outcome, patients lost to follow-up will be treated as censored observations. Since survival analysis methods, inherently account for censored data, no imputation will be applied to the primary outcome. This approach assumes that censoring is non-informative, meaning that the likelihood of being lost to follow-up is independent of the risk of the primary event.

3.9 Managing changes to the statistical analysis plan

Any modifications must be made prior to database unblinding and must be systematically validated by the scientific committee.

3.10 Software

All analyses will be performed using R software version 4.4.2.

4 References

[1] Latouche A, Porcher R, Chevret S, (2004) Sample size formula for proportional hazards modelling of competing risks. Stat Med 23: 3263-3274